A Better Hand

*The PIE Contrastive *-(t)ero-, the Numeral *du̯o-/du̯i-, and the Pre-PIE Hand-Noun*

Paper 2 of the der- project. Builds on Chavis & Claude (2026a).

[Working paper v10 — for scholarly review]

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Abstract

The PIE contrastive suffix *-(t)ero- has a well-documented oldest function — systematic spatial-direction marking across the IE family — and no satisfactory etymology. This paper proposes two convergent derivational accounts on the pre-PIE hand-root *de- of Paper 1. First, *-(t)ero- decomposes compositionally as *de- + *-r- + *-o-, where *de- is the independently attested directional particle and *-r- is the locative morpheme of Paper 3. Second, the numeral *du̯o-/du̯i- 'two' is reconstructed as *de- + *-u̯- + *-o/-i-, a parallel derivational sibling on the same root. The bilateral character of *-(t)ero- — its consistent function of marking 'one of a pair' — is then a morphological inheritance from the *-u̯- element of pairedness, not a typological consequence of body symmetry. Dunkel's (2014) published etymology unifying the prefixal *u̯i- and the numeral *du̯i-, together with his reconstruction of the inherited compound *u̯i-tero- (Vedic vítarám, OCS vŭtoru 'the second'), provides the principal external support. The OCP-driven reversal hypothesis is anchored empirically by Hittite kattera-, whose geminate -tt- identifies the deletion target in suffix-initial position. Probability: 35–45%.

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1. Introduction

The PIE contrastive suffix *-(t)ero- is one of the most productive morphemes in the inherited IE system, and one of the least explained. The standard handbooks agree on two points. First, its oldest documented function is spatial-direction marking: virtually every major PIE local particle forms a *-(t)ero- derivative naming the spatial contrastive of that particle (Latin citra/ultra/intra/extra/contra; Sanskrit ádhara-/uttara-/ápara-/ántara-; Hittite kattera-; and so on across the family). Second, the gradational 'more X' comparative function familiar from Classical Greek and Sanskrit is a late innovation confined to those two branches; the Handbook of Comparative and Historical Indo-European Linguistics is explicit that outside Greek and Indo-Iranian, *-(t)ero- and its bare alternant *-(e)ro- 'are confined to derivations of pronouns and particles.' What the handbooks do not provide is any account of where the suffix comes from, or why it should be the systematic spatial-direction marker applied to local particles across the family.

This paper proposes the missing etymology in two converging movements. The first derives *-(t)ero- compositionally as *de- + *-r- + *-o-, where *de- is the pre-PIE hand-element of Paper 1 and *-r- is the locative morpheme documented independently by Lipp (2009), Bauhaus (2019), and Kahl (2024). The second develops a parallel derivation on the same root: the numeral *du̯o-/du̯i- 'two' as *de- + *-u̯- + *-o/-i-, structurally parallel to *dei̯u̯-o- 'god'. The two derivations converge on the bilateral character of *-(t)ero-: if the suffix and the numeral 'two' are siblings on the same root, the 'one of a pair' meaning is a morphological inheritance rather than a typological accident.

The paper proceeds as follows. §2 documents the spatial-direction paradigm. §3 develops the bilateral character and the ambidexter argument. §4 sets out the compositional derivation and the t/d alternation. §5 develops the numeral branch and the morphological-identity argument, drawing on Dunkel's (2014) published evidence. §6 addresses the directionality question — whether *-(t)ero- is original or secondary — and the OCP mechanism. §7 presents the Hittite kattera- anchor. §8 reports the distributional test. §9 develops the Anatolian distribution argument. §10 addresses cross-linguistic typology. §11 sets out scope and remaining problems. §12 and §13 close with the remaining papers and the probability assessment.

2. The Spatial-Direction Paradigm

Table 1 presents the *-(t)ero- paradigm across the IE branches, restricted to inherited PIE-level formations. The table illustrates the systematic character of the spatial-direction function: for virtually every local particle that has a *-(t)ero- derivative in the inherited vocabulary, the derivative names the spatial contrastive of that particle.

PIE particle	Branch	*-(t)ero- form	Derived meaning
*ḱi- 'this, here'	Latin	*citer, citra*	'nearer, to this side'
ol-/uls- 'that, beyond'	Latin	*ulterior, ultra*	'farther, beyond'
*ud- 'up'	Sanskrit	*uttara-*	'upper, northern'
*ud- 'up, away'	Greek	*ὕστερος (hústeros)*	'later, after'
*ádh-/*n̥dh- 'below'	Sanskrit	*ádhara-*	'lower'
*ádh- 'below'	Avestan	*aδara-*	'lower'
*(s)up-/*uper- 'above'	Latin	*superus, super*	'upper, above'

PIE particle	**Branch**	***-(t)ero- form**	**Derived meaning**
*(s)up- 'above'	Av./OP	*upara-, upataro-*	'upper, on top of'
*ḱm̥t- 'below/alongside'	Hittite	*kattera-*	'lower; inferior' [§7]
*$h_1$en- 'in'	Latin	*inter, intra*	'between, within'
*$h_1$en- 'in'	Sanskrit	*ántara-*	'inner, between'
*$h_1$en- 'in'	Latin/Gr./Skt.	*intestina; ἔντερα; āntrá-*	'entrails' (lexicalized neuter)
*eks- 'out'	Latin	*exter, extra*	'outer, outside'
*kom- 'with'	Latin	*contra*	'opposite, against'
*pro- 'forward'	Av./OP	*fratara-*	'forward, more forward'
*pro- 'forward'	Greek	*πρότερος (próteros)*	'former, before'
*deḳs- 'right'	Latin	*dexter*	'right-handed'
*deḳs- 'right'	Greek	*δεξιτερός (deksiterós)*	'right (hand)'
*k^wo- 'who'	Latin / Skt. / Gr.	*uter; katára-; πότερος*	'which of two'
*al- 'other'	Latin	*alter*	'the other (of two)'
(deictic)	Greek	*ἕτερος (héteros)*	'the other'

*Table 1. Selected *-(t)ero- spatial-direction derivatives across the IE branches. Entries restricted to inherited PIE-level formations; bare *-(e)ro- alternants noted where relevant (e.g. Latin inferus). The lexicalization of the neuter of *$h_1$en-tero- as 'entrails' in three independent branches confirms the PIE-level age of the spatial adjective.*

Two features of the paradigm are relevant to the etymology proposed here. First, the spatial-direction function is systematic, not lexically idiosyncratic: it applies productively across a large and typologically diverse set of local particles in every major branch. Second, the bilateral character is consistent: every *-(t)ero- derivative names a spatial contrastive specifically in opposition to its implied counterpart, not merely a point on a scale. Latin citra 'on this side' presupposes ultra 'on that side'; Greek πρότερος presupposes ὕστερος; Latin dexter presupposes sinister. The suffix marks the selection of one member of a pair.

3. The Bilateral Character of *-(t)ero-

The standard handbooks agree that bilateral contrast is *-(t)ero-'s original function. Fortson's formulation is representative: the basic function was to 'set off a member of a pair contrastively from the other member of the pair, as in Greek skaiós 'left' vs. deksiterós 'right,' i.e. 'the one on the right (as opposed to the one on the left)" (Fortson 2010). Sihler (1995: 360–361) gives the same analysis. What the mainstream consensus does not provide is any account of where the bilateral character comes from.

The Latin compound ambidexter is the morphologically explicit witness for this function. It parses as ambi- (PIE *$h_2$mb^{h}i 'on both sides') + *deḱs- ('right') + -i- (linking vowel) + *-(t)ero-: 'having the right-side marker on both sides.' Festus's gloss confirms the parse: ambidexter qui dexteros ambo manus habet. The statement 'having the right-side marker on both sides' is semantically coherent only if *-(t)ero- normally marks one of two bilateral sides; ambi- then negates the contrast by applying the marker to both. The Greek cognate deksiterós, inherited independently from PIE, confirms that the bilateral frame is not a Latin formation but a PIE-level feature of the compound.

The framework offers two accounts of why *-(t)ero- should carry a bilateral frame. The first is typological: the locative *-r- anchors the directional *de- to the body as spatial frame of reference, and bodies have inherent bilateral symmetry. This is the framework's first-pass account and remains available if the stronger argument of §5 is rejected. The second and stronger account is morphological: the bilateral character is an inheritance from the same *-u̯- element of pairedness that produces the numeral 'two,' because *-(t)ero- and *du̯o- are derivational siblings on the same root.

4. The Compositional Derivation and the t/d Alternation

With the bare *de- directional particle (Paper 3) and the locative *-r- (Lipp 2009; Bauhaus 2019; Kahl 2024) both independently motivated, the compositional derivation of *-(t)ero- is direct. The suffix decomposes as *de- + *-r- + *-o- (thematic vowel). Each morpheme is independently attested with the right phonology and semantics. The combined *de-r-o- means 'at the place where the hand points/extends' — the side toward which one is oriented. The grammaticalization pathway HAND > SIDE is documented cross-linguistically in P'urhepecha, Mordvin, and several Mayan languages, where body-part terms function productively as relational nouns expressing spatial location (Heine & Kuteva 2002).

The t/d alternation needs to be addressed directly. The root is *de- with original d-, following the consistent d- vocalism of the lexical cognates documented in Paper 1. The t- of the grammaticalized suffix reflects a regular tendency toward devoicing of obstruents in suffix-initial position in the PIE morphological system: the inherited bound suffixes *-to-, *-ter-/-tor-, *-tro-, *-ti-, *-tu- are uniformly voiceless-initial. When *de- grammaticalized into a bound suffix, its initial d- devoiced to t- in conformity with this pattern. The *du̯- of *du̯o- preserves the original voicing because it is not in suffix-initial position: it is the surface root onset of a freestanding numeral, where the original d- is retained.

5. The Numeral Branch and the Morphological-Identity Argument

The central structural argument of the present paper is that *du̯o-/du̯i- ‘two’ and *-(t)ero- are derivational siblings on the same pre-PIE root, and that the bilateral character of *-(t)ero- is accordingly a morphological inheritance from the *-u̯- element of pairedness. The argument proceeds in four steps.

The Schrijver/Beekes observation. The starting point is a parenthetical remark in Schrijver (1995: 58), citing Beekes: ‘in the Greek forms, δ- might reflect *du- (cf. δώδεκα ‘12’ < *duŏ-)’ — that is, the δ- of Greek δῶρον might reflect the same *du- that surfaces in *duŏ- ‘two.’ The observation is uncommitted, but it registers the morphological connection the present paper develops.

The decomposition *du̯- as *de- + *-u̯-. The framework proposes that *du̯- itself decomposes as *de- + *-u̯-, with reduction of pre-tonic *de- to *d- before the *-u̯- suffix. The numeral *du̯o-/du̯i- is then a thematic derivative of the hand-element on the same pattern as *deiu̯-o- ‘god’ (Vedic devá-, Latin deus, Old Lithuanian deivas), which decomposes universally as *dei- + *-u̯- + *-o-. The structural parallel is direct: the same morphological operation (*Cei- + *-u̯- + *-o-) applied to different roots. The standard view treats *du̯- as primitive and unanalyzable; the framework proposes that it is decomposable, and that the same decomposition that explains *deiu̯-o- is available for it.

Dunkel’s published etymology. The framework’s morphological-identity claim receives its principal external support from Dunkel (2014, Lexikon der indogermanischen Partikeln, s.v. *u̯í). Dunkel takes the explicit position that the prefixal *u̯í- ‘apart, separated’ ‘arose by dissimilation from the compound first member *du̯i “two”’ in phonologically conditioned environments — specifically in compound-first-member position before stop-initial second elements. He grounds this dissimilation in a list of inherited PIE syntagms: *(d)u̯i $d^{h}eh_{1}$- ‘to halve, separate,’ *(d)u̯i der- ‘to tear apart,’ *(d)u̯i deḱ- ‘to deny.’ Three independent branches preserve the first of these as compositional verbs of dividing: Latin dīvidō (de Vaan 2008, s.v.: PIE *(d)ui-$d^{h}h_{1}$-), Sanskrit vi-dhā-, and Tocharian B wätk- (PIE *u̯i-$d^{h}h_{1}$-sḱe/o-; Malzahn 2010). Dunkel’s unification of *u̯i- and *du̯i- is his published position in the standard reference for IE particles, not a framework inference.

The inherited compound *u̯i-tero-. Dunkel’s entry on *u̯í- includes a sub-entry on the inherited compound *u̯i-tero- (*u̯i-, the dissimilated allomorph of *du̯i- ‘two,’ + *-(t)ero-, the contrastive suffix). The compound is reconstructed as inherited and survives across multiple branches: Vedic vítarám ‘further, sideways’; Avestan vítara- ‘sideways, later’; Old Church Slavonic vŭtoru ‘the second.’ The OCS reflex is decisive for the framework’s argument. *u̯i-tero- surfaces in OCS as the binary-ordinal numeral ‘the second one of two’ — a meaning that composes transparently from its constituents: ‘twoness’ (from *u̯i-) + ‘bilateral contrastive’ (from *-(t)ero-) = ‘the contrastive one of two’ = ‘the second.’ The bridge between the numeral and the contrastive suffix is provided by Dunkel’s published reconstruction; the framework’s specific contribution beyond Dunkel is the decomposition of *du̯- as *de- + *-u̯-.

Under the morphological-identity reading, the Latin compound ambidexter contains three reinforcing layers of pairedness-marking morphology: ambi- (*$h_2$mb^{h}i-, bilateral-marking adverb), *deḱs- (‘right [hand]’, from a lateralization domain), and *-(t)ero- (the contrastive suffix descending from the hand-root via *-u̯-). The compound’s synchronic meaning — ‘right-handed on both sides’ — explicitly negates a bilateral norm that *-(t)ero- encodes. Breton supplies an independent synchronic surface witness to the same architecture: daouarn ‘hands’ (= daou ‘two’ + dorn ‘hand’) marks the body-part dual with the pairedness morpheme, in a category Breton reserves exclusively for anatomically paired body parts.

6. The Directionality Question and the OCP Mechanism

The standard view holds that the bare suffix *-(e)ro- is original and that *-(t)ero- arose secondarily by metanalysis of t-final bases: when speakers reanalyzed the boundary of a form like *ḱm̥t-ero- as *ḱm̥t- + *-ero-, the t of the base became reattached to the suffix. This is the view of Wackernagel & Debrunner (1957) for Sanskrit, Schwyzer (1939) for Greek, and most comparative treatments. Bauhaus (2019) and Kahl (2024) offer the most recent elaborations, both arguing for a locatival origin of *-(e)ro-: they agree with the framework that *-r- is a locative morpheme, but hold that the t-bearing form is secondary.

The framework holds the reverse: *-(t)ero- is original, and bare *-(e)ro- forms arise by OCP-driven deletion of the suffix-initial *t- when the base ends in a stop. The phonological mechanism is the ‘Métron Rule’ of PIE cluster simplification at morpheme boundaries, documented by Byrd (2015) as a regular consequence of the Obligatory Contour Principle in PIE: when a base-final stop met a suffix-initial *t-, the resulting cluster was resolved by deletion of one of the two consonants. The key point is that Byrd’s own footnote 161 explicitly identifies suffix-initial deletion as a formally available alternative to root-final deletion: ‘if the */d/ of the root was lost, we would expect mé.trom; if the */t/ of the suffix was lost, we would expect mét.rom.’ Greek métron with single -t- is consistent with both outcomes. The framework proposes that suffix-initial deletion is the correct outcome in the *-(t)ero- formation specifically, and that this is confirmed by the Hittite test case.

On the question of what distinguishes the framework from Bauhaus’s and Kahl’s metanalysis accounts: both predict approximately the same surface distribution of bare and t-bearing forms.

The framework’s advantage is etymological rather than distributional: it explains why the t- was present in the first place (as the initial consonant of the hand-noun *de-, devoiced in suffix-initial position), while metanalysis takes the t- as given and explains only its propagation.

7. The Hittite Anchor: kattera-

The Hittite contrastive adjective kattera- ‘lower; inferior; further along’ (Kloekhorst 2008: 538–539) is the framework’s empirical anchor for the phonological argument. It is built on the ‘below/alongside’ particle *ḱm̥t-, the same element that gives Latin cum, Old Irish cét, Greek katá, and Hittite katta.

The derivation under the framework runs as follows: pre-PIE *ḱm̥t-tero- (root + suffix; dental cluster across morpheme boundary) undergoes OCP-driven simplification with suffix-initial deletion, giving PIE *ḱm̥t-ero-; *m̥ vocalizes to a and the *m before *t assimilates to geminate, giving Anatolian *katt-era-; regular Anatolian development yields Hittite kattera-.

The critical disambiguation is this: if the deletion had targeted the root-final *t instead of the suffix-initial *t, the surface form would be Hittite *katera- with a single intervocalic -t-, not the attested geminate -tt- of kattera-. The Hittite surface form selects suffix-initial deletion. This is the empirical content of Byrd’s footnote 161: there are surface forms that disambiguate the rule’s outcome, and kattera- is one of them.

8. The Distributional Test

Under the OCP-driven reversal hypothesis, bare *-(e)ro- forms arise by suffix-initial deletion when the base ends in a stop. The prediction is therefore that bare forms should cluster systematically on stop-final bases, while t-bearing *-(t)ero- forms should cluster on bases ending in vowels, sonorants, or fricatives. Table 2 reports the test applied to the inherited PIE positional adjective paradigm across four branches.

Branch	Bare *-(e)ro- on stop-final bases	t-bearing *-(t)ero- on non-stop bases
Sanskrit	ádhara- (*n̥dh-, dental asp.); ápara- (*ap-, labial)	uttara- (*ud-, dental; gemination)
Avestan	upara- (*up-, labial); aδara- (*adh-, dental)	fratara- (*pro-, vowel)
Latin	super (*(s)up-, labial); inferus (*n̥dh-, dental)	inter, intra (*en-, nasal); exter, extra (*eks-, fricative); ultra (*ol-, liquid); citra (*ḱi-, vowel); contra (*kom-, nasal)
Greek	ὕπερ (*(s)up-, labial)	πρότερος (*pro-, vowel); ἔντερα (*en-, nasal); δεξιτερός (*deḳi-, vowel)

*Table 2. The distributional test applied to the inherited PIE positional adjective paradigm. In approximately 19 of 19 etymologically transparent cases across the four branches, bare *-(e)ro- forms fall on stop-final bases and t-bearing *-(t)ero- forms fall on non-stop bases, as the OCP deletion hypothesis predicts.*

The Latin forms subter and propter are the clearest exceptions: both have *-(t)ero- on labial-stop-final bases. Lewis & Short and de Vaan (2008) treat them as Latin-internal formations modeled on existing *-(t)ero- words after the original phonological conditioning had ceased to be productive — which is what analogical extension without a live conditioning rule looks like. The pronominal possessives (noster, vester) are the test's acknowledged failure mode: they have *-(t)ero- on sibilant-final bases in Latin but bare *-(e)ro- in Gothic. The defense is that pronominal possessives constitute a small functional category that generalized branch-internally once the original conditioning rule had ceased to operate. It should be acknowledged that both the reversal hypothesis and the Bauhaus/Kahl metanalysis account predict approximately the same distribution; the test argues against the earlier 'irregular extension' framing, but does not by itself decide between the two principled accounts.

9. The Anatolian Distribution

Anatolian preserves exactly one *-(t)ero-/*-(e)ro- form productively: kattera-, a lexicalized bare-alternant relic (§7 above). No productive *-(t)ero- system exists in Hittite; the productive spatial-contrastive function is taken over by *-eti-Ho- (Hittite -ezziya-), seen in ḫantezziya- 'first, foremost' and appezziya- 'back, last.'

Under the standard view, this asymmetry is unexplained: Anatolian simply happens to have preserved only one member of the alternant pair and generalized a replacement for the productive function. Under the reversal hypothesis, the Anatolian distribution is a chronological prediction. If the productive *-(t)ero- system was still in the process of generalization at the time of the Anatolian split — as the Indo-Anatolian hypothesis maintains for various morphological categories — then Anatolian would be expected to preserve only what had already stabilized: lexicalized formations on the most frequently occurring local particles, where the OCP deletion had already produced bare forms. This is precisely kattera-. What Anatolian lacks is the productive extension of *-(t)ero- to new bases, which would have required a still-active morphological rule. Bauhaus (2019: §5) independently supports this chronological reading, observing that the directive *-o 'could keep its productivity at least until Anatolian,' while the locative *-r 'ceased to be productive in all attested languages' — exactly the asymmetry the framework predicts.

10. Cross-Linguistic Typology

The diachronic ordering BINARY-CONTRASTIVE > COMPARATIVE that the framework reconstructs for PIE *-(t)ero- has cross-linguistic support. Kahl (2024) documents parallel developments in two independent families. In Uralic, Proto-Uralic *-mpV is reflected in Finnish as the productive comparative, but preserves 'synchronically unusual binary-contrastive forms' — the relative pronoun jompi and interrogative kumpi 'which of two' — exactly the functional pairing the reversal hypothesis predicts for PIE. In Tungusic, Proto-Tungusic *-dımar functions both as an optional comparative marker and as a binary-contrastive marker, with Kahl noting that 'the binary-contrastive function is older and was secondarily drawn into comparative constructions' — the same diachronic order the framework proposes for PIE.

The direct pathway HAND > BINARY-CONTRASTIVE is not catalogued in Heine and Kuteva (2002), but its component links are documented: HAND > SIDE in P'urhepecha, Mordvin, and Mayan, and SPATIAL > BINARY-CONTRASTIVE in Tungusic and Uralic. The English idiom 'on the one hand … on the other hand' preserves the direct use of HAND as a binary-contrast pivot in living language; the function it performs is exactly the function the framework claims for pre-PIE *de- in its grammaticalized form.

11. Scope and Remaining Problems

The directionality question remains genuinely contested. The reversal hypothesis and Bauhaus's and Kahl's metanalysis accounts are both consistent with the distributional evidence; the framework's advantage over metanalysis is etymological rather than distributional. The Hittite kattera- anchor is the strongest evidence for suffix-initial deletion specifically, but a reviewer committed to metanalysis could accommodate it without abandoning that framework.

The grammaticalization pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE has typological support in its component links but is not directly attested as a complete chain in any language. The morphological-identity argument of §5 provides the stronger motivation for the bilateral character; the typological argument is a supporting parallel, not a demonstration.

The decomposition of *du̯- as *de- + *-u̯- is the bolder claim of the paper. The standard view treats *du̯- as primitive; the framework decomposes it. The closest structural parallel is *deiu̯-o- 'god,' and Dunkel's published unification of *u̯i- and *du̯i- provides indirect support. Whether this is sufficient to carry the decomposition is a judgment for specialists in IE morphology.

The kinship suffix *-(h_2)ter- (in *māter-, *p$h_2$ter-, *b^hrāter-, *d^hug$h_2$ter-) shares the t-r structure of *-(t)ero-, and Latin matertera 'maternal aunt' suggests a Latin-internal extension of *-(t)ero- to a kinship term. Whether the kinship suffix descends from the same pre-PIE etymon is a question the framework raises but does not address.

12. The Remaining Papers

Paper 3 ('The Bare Hand') establishes *de- as an independently attested PIE directional particle and decomposes the give-verb *deh$_3$- as *de- + verbal extension *-h$_3$-. The discriminating evidence is a class-by-class asymmetry in the Greek root-aorist 2pl imperative paradigm: roots the framework analyzes as bimorphemic preserve a bare-CV form in the 2pl (*deh$_3$- → δότε; *d^{h}eh$_1$- → θέτε), while roots the framework does not analyze as bimorphemic preserve the laryngeal-conditioned long vowel (*steh$_2$- → στῆτε; *ĝn̥h$_3$- → γνῶτε). The asymmetry is established in the standard grammatical tradition; the framework provides a structural explanation for it.

Paper 4 ('The Hand That Takes') carries the framework into Anatolian, deriving Hittite dā- 'to take' from a pre-Anatolian directive univerbation *de + a — an instance of the Hittite Set 1 directional system in which a local element follows the directional particle *-a to contract into a fixed form. The take/give polarity in Anatolian — the fact that Anatolian's reflex of *deh$_3$- means 'take' while post-Anatolian reflexes mean 'give' — resolves as a structural consequence of the directional element, and receives independent semantic support from Norbruis (2021).

13. Probability Assessment

Conditional on Paper 1 being correct, we assess the probability of the central thesis of this paper — that *-(t)ero- decomposes compositionally as *de- + *-r- + *-o-, with *du̯o-/'two' as a derivational sibling — at approximately 35–45%.

In favour: the locative status of *-r- now rests on three independent published treatments from distinct scholarly traditions; the compositional semantics are transparent and motivated by documented cross-linguistic pathways; Dunkel's (2014) published evidence — the *u̯i- ← *du̯i- dissimilation, the reconstructed *u̯i-tero- compound, and the OCS vŭtoru 'the second' attestation — is the strongest external support the framework has at any stratum; and the Hittite kattera- anchor gives the phonological argument empirical bite that purely distributional arguments lack. Against: the decomposition of *du̯- as *de- + *-u̯- is non-standard and rests primarily on the structural parallel with *deiu̯-o-; the directionality question is not settled by the distributional evidence alone; and the grammaticalization chain is reconstructed rather than directly attested.

The marginal probability for the four-paper project as a whole — integrating conditional probabilities across Papers 1–4 — is in the range 1–4%, with much of the evidence supporting multiple strata simultaneously. We regard the central claim as more likely wrong than right at that level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

Authorship and Contributions

Chavis's contributions. The hypothesis that *der- 'hand' underlies the contrastive *-(t)ero- was first articulated by Chavis, as was the slogan 'the ter in *-tero- is hand.' Chavis directed the research agenda at every stage: formulating the hypothesis that ambidexter is the morphologically transparent compound witness for the bilateral character; posing the question 'whence comes the r in *-tero-?' that opened the compositional analysis of Paper 3; directing the development of the numeral branch *du̯o-/du̯i- as a parallel derivational outcome; identifying the Schrijver/Beekes δ- ← *du- observation as the seed; and directing the framing of *du̯- as *de- + *-u̯- paralleling *deiu̯-o- 'god.' All fundamental decisions about scope, framing, and direction rest with Chavis, who also supplies all intellectual continuity across sessions through systematic transcript management.

Claude's contributions. Claude reviewed Dunkel's *u̯í- entry directly from the source, identifying the *u̯í- ← *du̯i- dissimilation, the *u̯i-tero- sub-entry with branch attestations, and the syntagm list grounding the dissimilation rule. Claude drafted the §5 argument sequence and integrated it with the empirical apparatus of §§6–9; identified the structural parallel with *deiu̯-o- 'god' as the closest morphological analog for the *du̯- decomposition; developed the Anatolian distribution argument (§9); proposed the structuring of the control-root argument in §7; and surfaced the Tungusic and Uralic typological parallels. Claude drafted all prose sections from Chavis's structural and substantive direction. The probability estimate in §13 is Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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